BACKTRACK: pitch and Q&A

One sentence

BACKTRACK helps a learner find a missing skill, practice it with Khan Academy, and return to the lesson they need now.

Ten seconds

A student is stuck but doesn't know what to review. BACKTRACK finds a useful earlier step, guides Khan practice, and checks the return.

Thirty seconds

A learner knows which question is stopping them, but may not know which earlier skill needs work. BACKTRACK uses their answer to choose a starting check, offers focused Khan practice, and brings them back to a fresh problem. Two wrong answers can lead to different help. The proposed school pilot will measure who gets back to the lesson and can still do it 7-14 days later.

Five-minute pitch script

Use this as a rehearsal script, not text to read from the slides. Leave time for pauses and a brief interaction. The 15-slide PDF can explain the full proposal when a live demonstration is impractical.

"A student gets stuck on today's math and is told to review. They can find a whole lesson, but which earlier step actually needs work? BACKTRACK makes that decision easier.

The idea is a GPS for learning. The destination is the lesson the student needs now. The route goes back to a useful earlier skill, gives that skill some focused practice, then returns to the destination.

Consider this quadratic equation. The difficulty may come from finding factors, multiplying brackets, or turning factors into solutions. A broad search for quadratics does not tell the learner which of those steps to work on first.

BACKTRACK begins with the task and the learner's answer. Imagine two answers: three and four, or two and six. Both are wrong as solutions to this equation, but they give different clues. The first pair fits the factors and points us toward signs and solutions. The second points us toward checking the factor pair.

Those answers lead to different starting checks. The route explains the choice. When the learner demonstrates a step, its review can leave the route. When a step needs work, BACKTRACK opens a small learning stop: one idea, a worked example, and a chance to try it.

Khan Academy supplies the learning resources. The factoring stop includes a ninety-four-second segment about matching a sum and product, with an option to continue watching. The learner can also work through the concept and practice directly inside BACKTRACK. Fresh problems then check whether they can return to the goal.

Slide seven invites you to try it. Click the button or scan the code, compare the two wrong answers, and see why they get different help. The five sample destinations include quadratics, brackets, fractions, ratios, and reading graphs.

The school question is simple: can learners who were stuck complete a fresh task after practice, and still do it a week or two later? We call that retained reentry. We will report the return among learners who began unable to complete the goal, including those who do not finish.

From November 2026 through March 2027, we propose eighty to one hundred twenty learners across two or three cohorts. Teachers will confirm the destinations and assessments, share the relevant route, and assign the matched practice in their existing Khan classes. The routine fits recurring study, remediation, or laboratory time.

We will compare the BACKTRACK-supported routine with teacher-selected Khan practice using similar time and support. Both will receive fresh assessments after practice and again seven to fourteen days later. We will also record repeated Khan use, learner frustration, teacher effort, and actual costs.

Progress gives learners a reason to continue. A checked step removes unnecessary review. A repair moves them closer to the goal. A later visit preserves earlier work. Short sessions can pause without forcing a restart.

The same routine can grow through a destination kit: the goal, reviewed skill map, checks, matched Khan material, and facilitator guide. Another educator should be able to run that kit and tell us where it needs improvement.

Our one-hundred-thousand-peso implementation scenario prioritizes learning design, access, school delivery, and evaluation. Learner access stays free. Measured costs and conversations with institutional budget owners will guide continuation.

We are the University of the Philippines Manila BACKTRACK team. We want students to leave a stuck moment with a useful next step, and a clear route back to today's lesson."

Demonstration rehearsal

Open https://backtrack-learning.vercel.app/demo before the slot. Tap 3 and 4, then open the repair: the route starts with signs and solutions. Replay and try 2 and 6: it starts with factor pairs. Show the local concept or focused Khan segment, then point to the fresh return. Use the static deck for later states if the slot is short. Present the interaction as the sample it is.

Hostile questions and answers

Isn't this just adaptive learning? Yes, it belongs to that family. The proposed contribution is a specific Khan-centered recovery routine, including a current-class return outcome. We must demonstrate value beyond a simpler routine.

Doesn't Khan already personalize? It does. Learning Paths and mastery tools are real. We start from a selected current destination and test the need for this lightweight entry point. We should not deploy BACKTRACK where an existing Khan workflow serves the school better.

Why not Gemini? Gemini can diagnose and plan study. Our difference is a reviewed, inspectable route and school implementation around Khan practice, with a formal reentry test. That is a hypothesis about useful delivery, not a claim that Gemini cannot imitate a feature.

Why Khan rather than any video site? The proposed school routine uses coherent explanations, exercises, assignments, and teacher reports in one existing learning environment. A video player alone would not satisfy our integration claim.

Where does Khan data come from? The public experience records resource opens and learner reports locally. It does not receive Khan results. In the pilot, authorized teachers will use available assignment reports/exports and reconcile them with pseudonymous records. Availability must be verified before enrollment.

How do you know the gap caused the difficulty? We do not establish causation from a wrong answer. It suggests a prerequisite to investigate. Targeted checks, later performance, teacher review, and a controlled comparison strengthen the evidence. The wording must remain "suspected blocker" until justified.

What does shortest mean? A small useful evidence-supported route to the chosen destination. We do not solve a globally optimal shortest-path problem or guarantee minimum time.

What if confidence is wrong? Correctness is decided by mathematics. Confidence changes follow-up or support. It never grants mastery or changes the score. We will evaluate whether the extra question is useful enough to keep.

What if they use AI to cheat? Remote practice cannot reliably establish who produced an answer. The interface explains that misleading answers produce an unhelpful route. Formal outcome tasks are separately supervised. We do not claim cheating detection.

Does two correct answers prove mastery? No. It supports a limited route decision. Formal reentry uses reviewed fresh tasks and a delayed check. Broader and longer-term mastery requires more evidence.

Who builds the graph? The team authors destination kits and educators review dependencies, discriminating questions, alternate methods, and curriculum fit. Versioning records changes. Content review is a real scaling cost.

How does it know today's lesson? A teacher or learner selects the destination. The public tool does not read an LMS or infer a class schedule automatically.

Why another tool for teachers? It must earn its place. A teacher shares a destination and uses an existing Khan class. The pilot measures the new setup/reporting burden and compares with teacher-selected practice. If it costs more effort without useful gains, simplify or stop that workflow.

What about no internet? Local progress can persist and the open page can continue. Khan content and initial loading still need a connection. Use scheduled access or local worked examples and record excluded/delayed learners. There is no full offline Khan claim.

What if the gap takes hours? The route can pause and resume. Short sessions do not erase deep prerequisites. Repeated difficulty leads to human support, not a guaranteed quick fix.

How will you establish impact? With baseline eligibility, fresh post tasks, delayed tasks, a clear denominator, and a feasible comparison. Public engagement and unsupervised checks are not added to formal school outcomes.

Who pays after March? That is unvalidated. We will measure actual costs and interview institutional budget owners in university extension, schools, nonprofits, LGUs, and CSR. We will not claim an invented package price proves sustainability.

How do you protect minors? School approval, required parent/guardian permissions, adult oversight, minimal data, school-controlled identity keys, restricted records, aggregate reporting, and an institutional review process. We do not claim blanket legal compliance from the software design.

Why does this deserve attention? A learner can have access to excellent lessons and still be unable to choose useful help. BACKTRACK makes that decision visible and testable: the mistake informs a check, the repair fits the step, and the learner returns to the task. The opportunity is to make existing Khan resources easier to use at the moment of difficulty.

Weak answers to avoid

"Nobody has done this." "Our AI finds the exact gap." "A video open proves meaningful use." "Two correct answers establish mastery." "We will reach thousands." "Schools will pay PHP 25,000." "Our design makes learning addictive." "We already partner with Khan." None of these is established.